Exercise #2 Geomechanics and Structural Geology - 18 February 2026 (Part of Prof. Giovanni Bertotti)

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Task 1

Provide a matrix and a Mohr diagram representation of the following states of stress:

- Hydrostatic compression at a water depth of 3 km
- Uniaxial tension (σ_3 = -5 MPa)
- Triaxial compression (σ_3 = 5 MPa; σ_2 = 8 MPa; σ_1 = 10 MPa)
- Tension and compression (σ_3 = -3 MPa; σ_2 = 0 MPa; σ_1 = 10 MPa)

Solution:

For this task, the assumptions being used are:

- Compression is positive and tension is negative.
- Principal stress order is $\sigma_1 \geq \sigma_2 \geq \sigma_3$.
- For hydrostatic water load at depth $z = 3$ km:

$$p = \rho_w g z = 1000 \times 9.81 \times 3000 = 29.43 \text{ MPa}$$

- Stress tensor in principal coordinates is:

$$[\sigma] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

- Because each stress state is given with three principal stresses $(\sigma_1, \sigma_2, \sigma_3)$, so it is a 3D state of stress. Mohr circles in 3D are built from the pairs (σ_1, σ_2), (σ_2, σ_3), and (σ_1, σ_3).

Direct formulas used:

- Circle (σ_1, σ_2):

$$C_{12} = \frac{\sigma_1 + \sigma_2}{2}, \quad R_{12} = \frac{|\sigma_1 - \sigma_2|}{2}$$

- Circle (σ_2, σ_3):

$$C_{23} = \frac{\sigma_2 + \sigma_3}{2}, \quad R_{23} = \frac{|\sigma_2 - \sigma_3|}{2}$$

- Circle (σ_1, σ_3):

$$C_{13} = \frac{\sigma_1 + \sigma_3}{2}, \quad R_{13} = \frac{|\sigma_1 - \sigma_3|}{2}$$

C is the center of a Mohr circle, R is the radius of a Mohr circle (equal to the maximum shear stress for that pair).

Case 1: Hydrostatic compression (3 km water depth)

Principal stresses:

$$\sigma_1 = \sigma_2 = \sigma_3 = 29.43 \text{ MPa}$$

Stress matrix:

$$[\sigma]_A = \begin{bmatrix} 29.43 & 0 & 0 \\ 0 & 29.43 & 0 \\ 0 & 0 & 29.43 \end{bmatrix} \text{ MPa}$$

Mohr representation: all circles collapse to one point ($R = 0$).

```
Case 1 = Hydrostatic compression (3 km water depth)
sigma_1=29.43 MPa, sigma_2=29.43 MPa, sigma_3=29.43 MPa
Stress matrix [MPa]:
[[29.43  0.    0.  ]
 [ 0.   29.43  0.  ]
 [ 0.    0.   29.43]]
Mohr circles (C, R) [MPa]:
  (sigma_1, sigma_3): C=29.43, R=0.00
  (sigma_1, sigma_2): C=29.43, R=0.00
  (sigma_2, sigma_3): C=29.43, R=0.00
Plot Mohr: Case 1 = Hydrostatic compression (3 km water depth)
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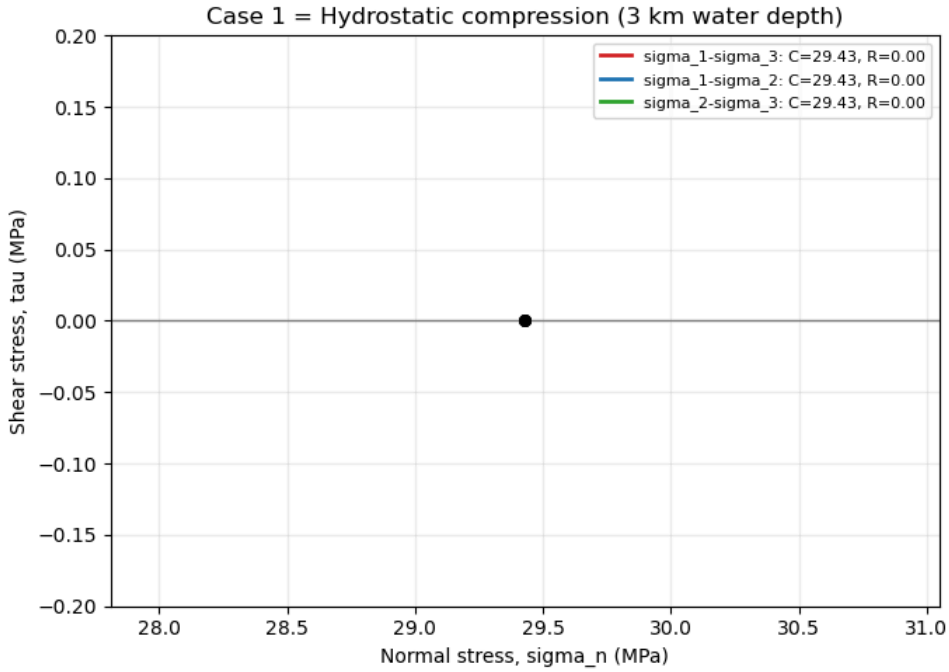


Case 2: Uniaxial tension

Principal stresses:

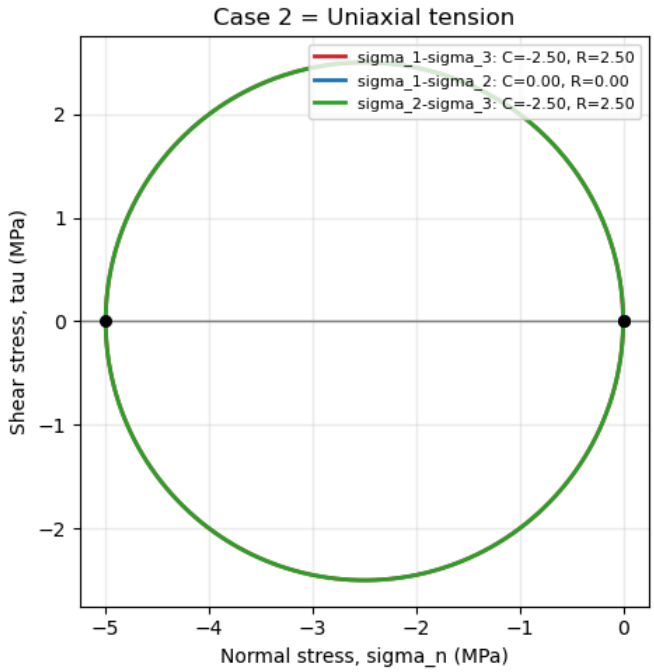
$$\sigma_1 = 0\ \mathrm{MPa}, \quad \sigma_2 = 0\ \mathrm{MPa}, \quad \sigma_3 = -5\ \mathrm{MPa}$$

Stress matrix:

$$[\sigma]_B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -5 \end{bmatrix}\ \mathrm{MPa}$$

Mohr representation: one non-zero circle between σ_1 and σ_3 (same as σ_2 and σ_3 because $\sigma_1 = \sigma_2$).

```
Case 2 = Uniaxial tension
sigma_1=0.00 MPa, sigma_2=0.00 MPa, sigma_3=-5.00 MPa
Stress matrix [MPa]:
[[ 0.  0.  0.]
 [ 0.  0.  0.]
 [ 0.  0. -5.]]
Mohr circles (C, R) [MPa]:
  (sigma_1, sigma_3): C=-2.50, R=2.50
  (sigma_1, sigma_2): C=0.00, R=0.00
  (sigma_2, sigma_3): C=-2.50, R=2.50
Plot Mohr: Case 2 = Uniaxial tension
```



Case 3: Triaxial compression

Principal stresses:

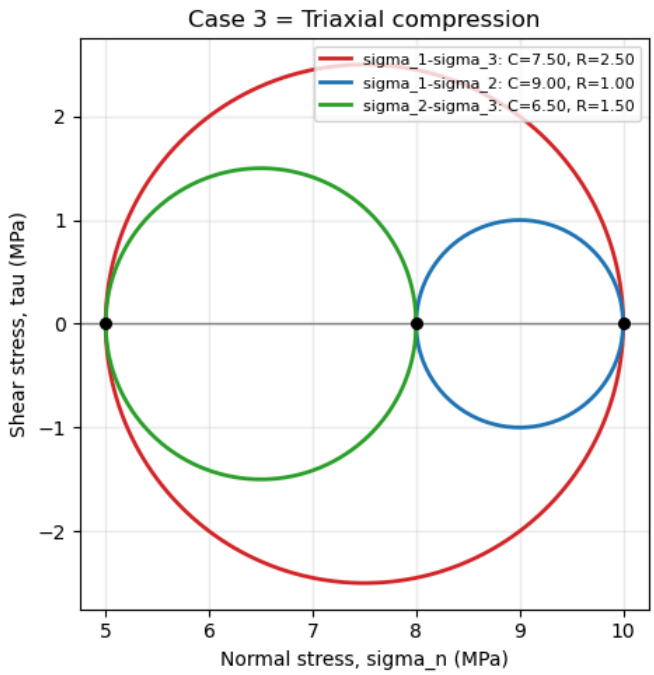
$$\sigma_1 = 10\ \mathrm{MPa}, \quad \sigma_2 = 8\ \mathrm{MPa}, \quad \sigma_3 = 5\ \mathrm{MPa}$$

Stress matrix:

$$[\sigma]_C = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 5 \end{bmatrix}\ \mathrm{MPa}$$

Mohr representation: three distinct circles in the compressive stress domain.

```
Case 3 = Triaxial compression
sigma_1=10.00 MPa, sigma_2=8.00 MPa, sigma_3=5.00 MPa
Stress matrix [MPa]:
[[10.  0.  0.]
 [ 0.  8.  0.]
 [ 0.  0.  5.]]
Mohr circles (C, R) [MPa]:
  (sigma_1, sigma_3): C=7.50, R=2.50
  (sigma_1, sigma_2): C=9.00, R=1.00
  (sigma_2, sigma_3): C=6.50, R=1.50
Plot Mohr: Case 3 = Triaxial compression
```



Case 4: Tension and compression

Principal stresses:

$$\sigma_1 = 10\ \text{MPa}, \quad \sigma_2 = 0\ \text{MPa}, \quad \sigma_3 = -3\ \text{MPa}$$

Stress matrix:

$$[\sigma]_D = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -3 \end{bmatrix} \text{MPa}$$

Mohr representation: mixed compression-tension state, with circles spanning negative and positive normal stress.

```
Case 4 = Tension and compression
sigma_1=10.00 MPa, sigma_2=0.00 MPa, sigma_3=-3.00 MPa
Stress matrix [MPa]:
[[10.  0.  0.]
 [ 0.  0.  0.]
 [ 0.  0. -3.]]
Mohr circles (C, R) [MPa]:
  (sigma_1, sigma_3): C=3.50, R=6.50
  (sigma_1, sigma_2): C=5.00, R=5.00
  (sigma_2, sigma_3): C=-1.50, R=1.50
Plot Mohr: Case 4 = Tension and compression
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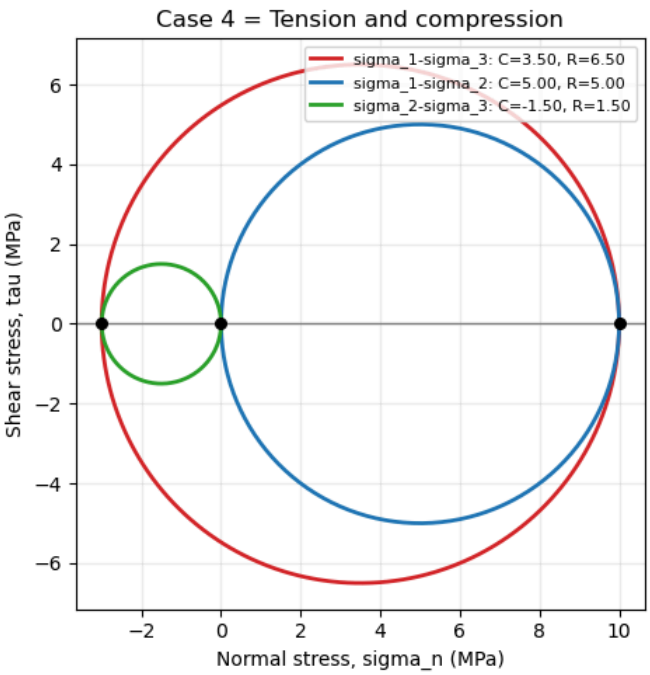

Interpretations:

- Case 1 (hydrostatic compression): all principal stresses are equal, so all three Mohr circles collapse to one point at $\sigma_n = 29.43$ MPa and $\tau = 0$.
- Case 2 (uniaxial tension): one stress is tensile ($\sigma_3 = -5$ MPa), producing circles that cross into negative normal stress domain.
- Case 3 (triaxial compression): all stresses are compressive, producing three distinct circles entirely in positive σ_n range.
- Case 4 (tension + compression): mixed stress state gives the largest spread, from negative to positive normal stress, with the largest circle from σ_1 to σ_3.

Task 2

Using the (2D) Mohr diagram for stress, determine the normal and shear stresses acting on a plane (line in 2D) oriented N30.

Boundary conditions are:

- σ_1 is oriented N-S and has a magnitude of 13 MPa.
- σ_2 has a magnitude of 7 MPa.

Solution:

Given:

- $\sigma_1 = 13$ MPa (oriented N-S).
- $\sigma_2 = 7$ MPa (perpendicular to σ_1).

- Plane of interest: oriented N30, so $\theta = 30^\circ$ relative to N-S axis.

Use stress transformation equations:

$$\sigma_n = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2}\cos(2\theta)$$

$$\tau = \frac{\sigma_1 - \sigma_2}{2}\sin(2\theta)$$

Since $\theta = 30^\circ$, then:

$$2\theta = 60^\circ$$

Normal stress:

$$\sigma_n = \frac{13 + 7}{2} + \frac{13 - 7}{2} \cdot \cos(60^\circ)$$

$$\sigma_n = 10 + 1.5 = 11.5 \text{ MPa}$$

Shear stress:

$$\tau = \frac{13 - 7}{2} \cdot \sin(60^\circ)$$

$$\tau \approx 2.598 \approx 2.6 \text{ MPa}$$

The Mohr circle is shown as follow:

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Normal stress, sigma_n = 11.50 MPa
Shear stress, tau = 2.60 MPa
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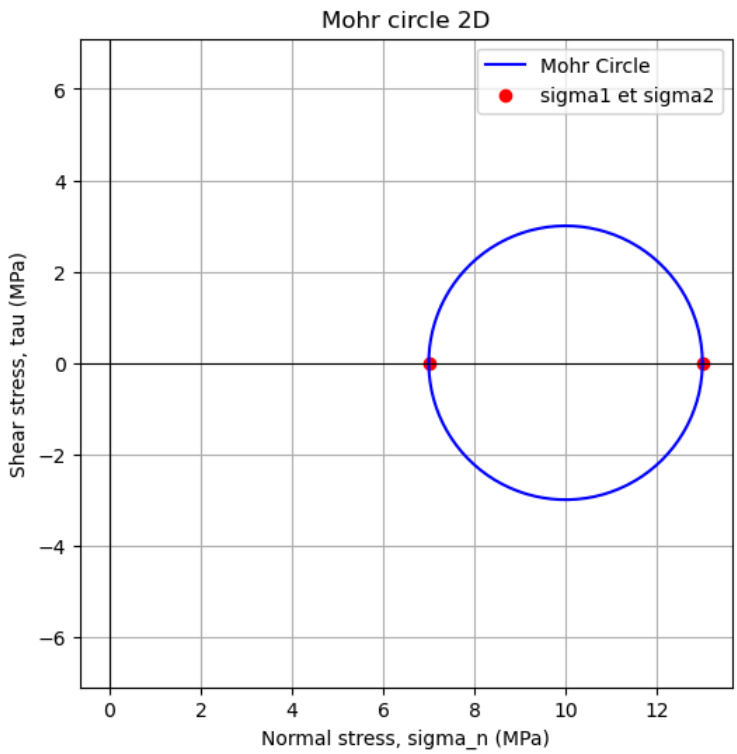


Task 3

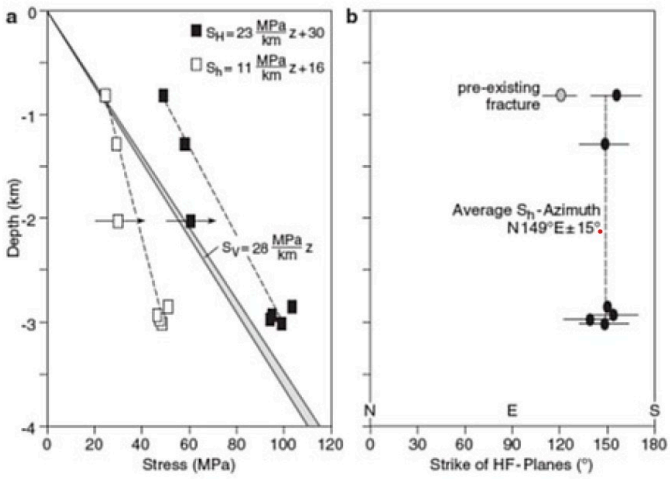
Stress changes in the KTB borehole (plots on the below):

a) Draw Mohr diagrams to describe the state of stress of rocks at 0.5, 2.5, and 4 km (extrapolate curves when needed).

b) On the same graph, plot a deviatoric stress curve.

c) Using the second plot, determine the position in space of the three principal stresses.

d) Predict the stress regime from surface to a depth of 4 km.



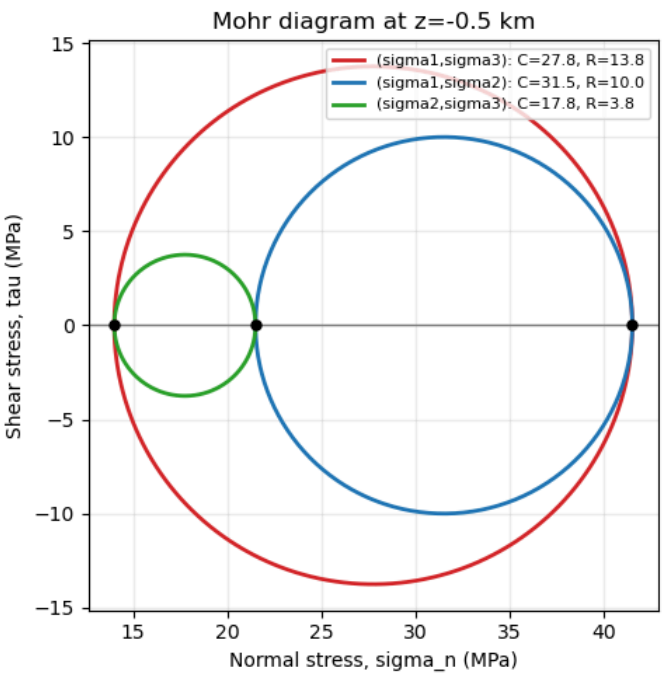
Solution:

The left plot: showing stress magnitude vs depth for the KTB borehole. The three curves represent the vertical stress S_V, maximum horizontal stress S_H, and minimum horizontal stress S_h as functions of depth. From the trend lines in the provided KTB plot, we use linear approximations (MPa, depth in km): $S_V(z) = 28z$; $S_H(z) = 23z + 30$; and $S_h(z) = 11z + 16$.

The right plot: displaying the orientation (azimuth) of the minimum horizontal stress S_h vs depth. This shows that S_h is approximately oriented at N149°E ($\pm 15^\circ$), which helps determine the spatial position of principal stresses in the field.

(a) Mohr Diagrams of rocks at 0.5, 2.5 and 4 kms

Mohr diagram at z = -0.5 km



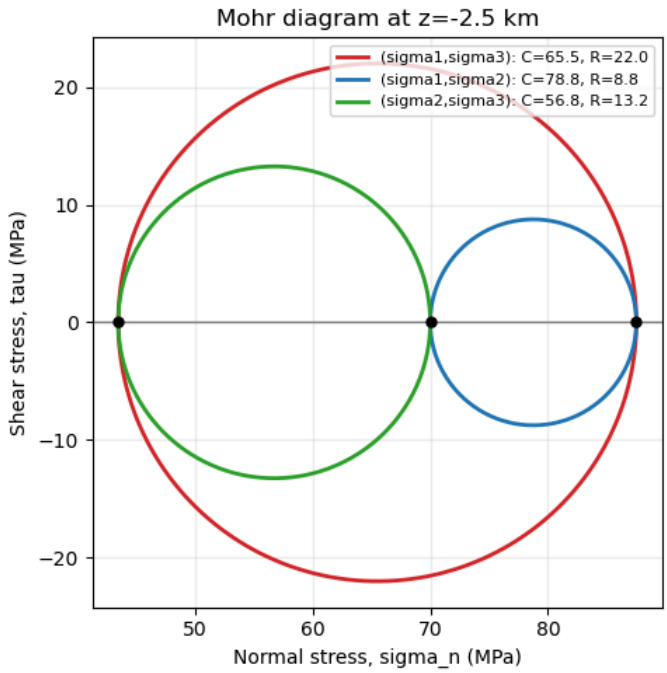
At depth 0.5 km, we can get information:

- $\sigma_H = 23(0.5) + 30 = 41.5$ MPa
- $\sigma_V = 28(0.5) = 14.0$ MPa
- $\sigma_h = 11(0.5) + 16 = 21.5$ MPa

Order of stresses at this depth: $\sigma_H > \sigma_h > \sigma_V$, so $\sigma_1 = 41.5$ MPa, $\sigma_2 = 21.5$ MPa, $\sigma_3 = 14.0$ MPa

So, *deviatoric stress:* $q = \sigma_1 - \sigma_3 = 41.5 - 14.0 = 27.5$ MPa.

Mohr diagram at z = -2.5 km



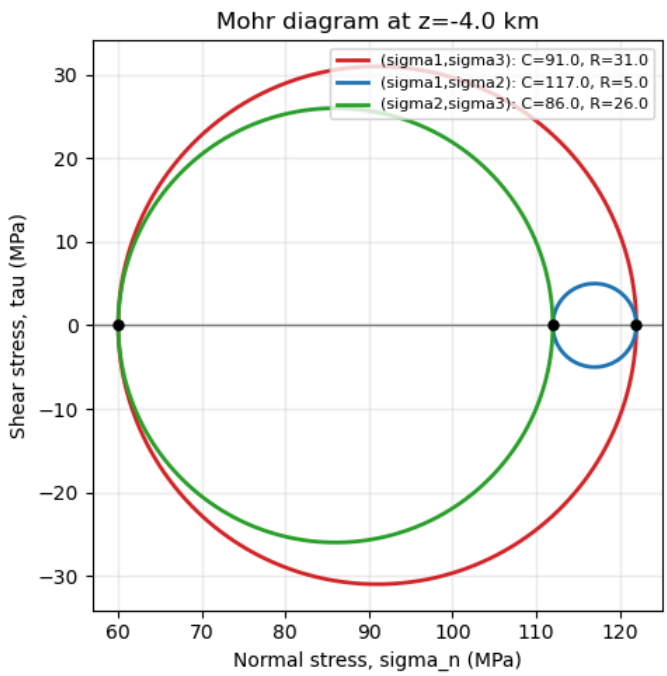
At depth 2.5 km, we found that:

- $\sigma_H = 23(2.5) + 30 = 87.5$ MPa
- $\sigma_V = 28(2.5) = 70.0$ MPa
- $\sigma_h = 11(2.5) + 16 = 43.5$ MPa

Order of stresses at this depth: $\sigma_H > \sigma_V > \sigma_h$, so $\sigma_1 = 87.5$ MPa, $\sigma_2 = 70.0$ MPa, $\sigma_3 = 43.5$ MPa

So, the *deviatoric stress*: $q = \sigma_1 - \sigma_3 = 87.5 - 43.5 = 44.0$ MPa (positive, larger than at 0.5 km)

Mohr diagram at z = -4.0 km



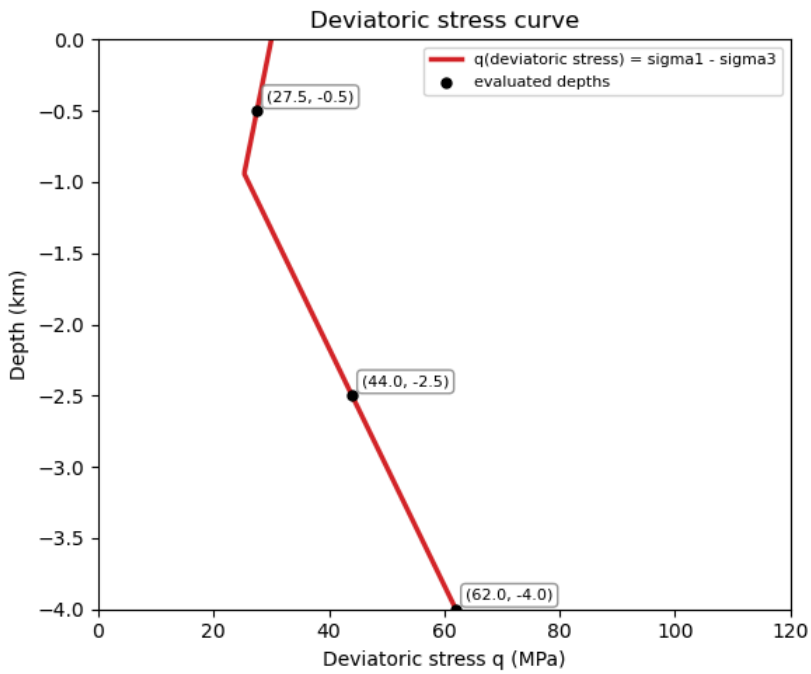
At depth 4.0 km, we found:

- $\sigma_H = 23(4.0) + 30 = 122.0$ MPa
- $\sigma_V = 28(4.0) = 112.0$ MPa
- $\sigma_h = 11(4.0) + 16 = 60.0$ MPa

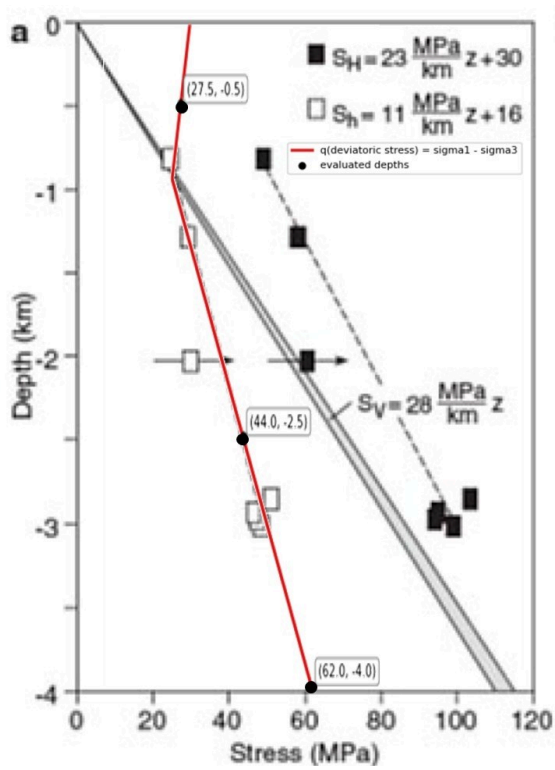
Order of stresses at this depth: $\sigma_H > \sigma_V > \sigma_h$, so $\sigma_1 = 122.0$ MPa, $\sigma_2 = 112.0$ MPa, $\sigma_3 = 60.0$ MPa

Therefore, *deviatoric stress*: $q = \sigma_1 - \sigma_3 = 122.0 - 60.0 = 62.0$ MPa (positive, largest among the three depths)

(b) Deviatoric Stress Curve (0-4 km)



Completing the deviatoric plot in the above graph, another version of a plot with overlay of KTB borehole stress vs depth is :



(c) Determining The Position in Space of The Three Principal Stresses (Using Graph b)

From the figure provided (plot b) in the task, S_h is horizontal with average azimuth N149°E ($\pm 15^\circ$). S_H is horizontal and perpendicular to S_h (approximately N59°E or N239°E). S_V is vertical.

(d) Predicting The Stress Regime from The Surface to A Depth of 4 km

To determine stress regime, we need to identify the crossover depth (that we can see it in graph *a* in the task) where $S_V = S_h$:

$$28z = 11z + 16$$

$$z = 0.94 \text{ km}$$

This creates two stress regimes:

Regime 1: Shallower interval (0 - 0.94 km)

- Stress ordering: $S_H > S_h > S_V$
- Principal stresses: $\sigma_1 = S_H$ (horizontal), $\sigma_2 = S_h$ (horizontal), $\sigma_3 = S_V$ (vertical)
- Since horizontal stress is the largest while vertical stress is the smallest, this zone is strike-slip faulting regime.

Regime 2: Deeper interval (0.94 - 4 km)

- Stress ordering: $S_H > S_V > S_h$
- Principal stresses: $\sigma_1 = S_H$ (horizontal), $\sigma_2 = S_V$ (vertical), $\sigma_3 = S_h$ (horizontal)
- Since horizontal stress is the largest while vertical stress is in between, this zone is strike-slip faulting regime with an obqlique possible movement.

Task 4

Spanish Peak stress field(s)

The Spanish peak volcanic system is composed of an intrusive "plug" and a large number of dykes. As you see from the map, the pattern of these dykes is not random.

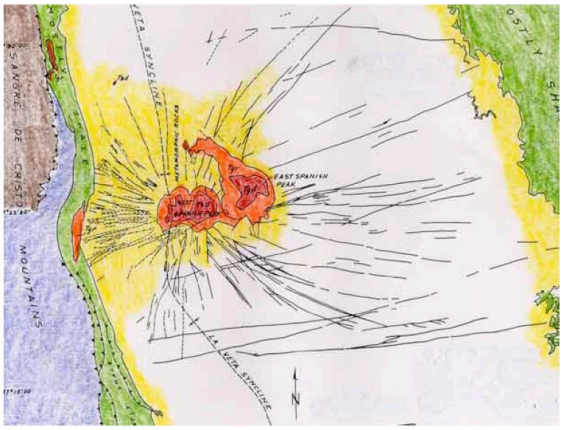
1. What are the relations between dykes and principal stresses?

2. What is the minimum number of stress fields which you can define? Explain your answer.

3. Draw the stress trajectories associated with the stress fields you have defined (use the map on the following page).

4. Define the tectonic origin of the stress field(s).

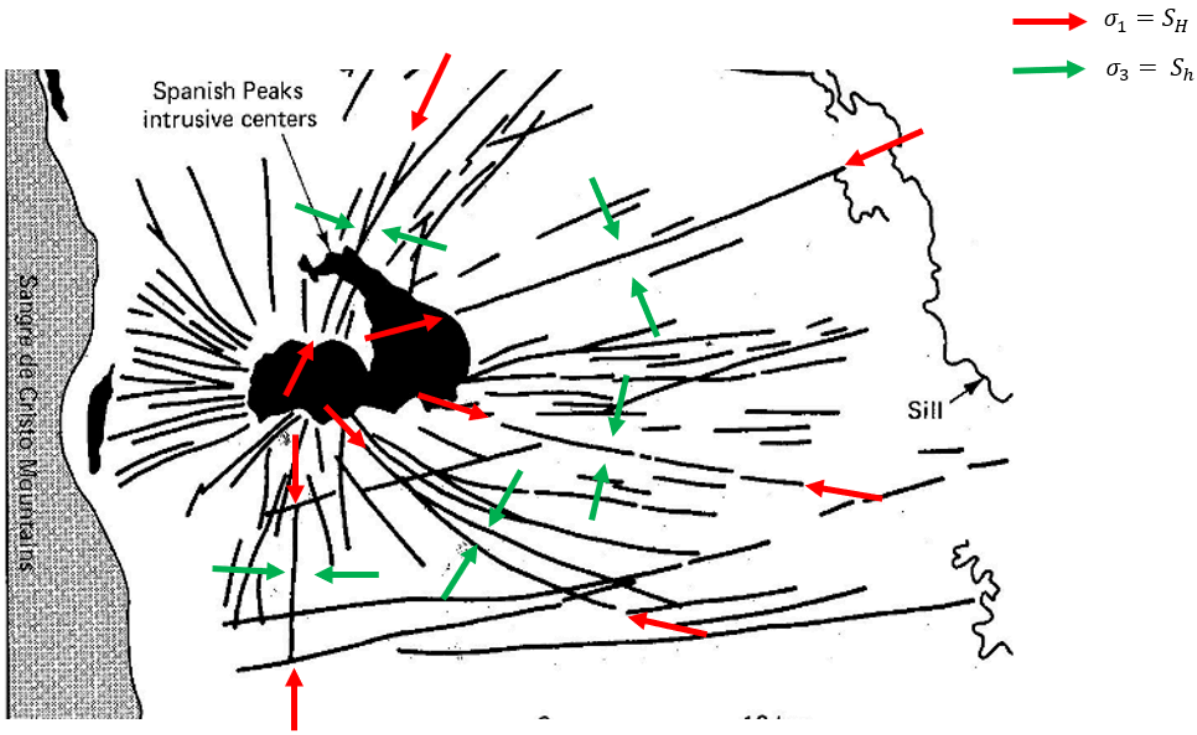
5. Make a geologic history of the area.



Solution:

1. Dykes will grow parallel to the σ_1 direction. As, the magmatic plug (the red colour part in the map) has such strong confinement that was strong enough to retain any upcoming pressure from the bottom. Therefore, the dykes intrusion was transmitted to fractures which likely occured parallel to σ_1, which also acted as maximum horizontal stress.

2. The map shows different dyke orientations, with a radial orientation and uniform nearly parallel orientation (over long distances). Therefore, we have to define at least two stress fields. The first stress field causing the radial oriented dykes, due to very local upcoming pressure from the bottom. The second stress field causes the nearly parallel oriented dykes over large distances.

3. 

4. The prominent set of radial dikes seen in the picture is focused on the West Spanish Peak. This is our first tectonic origin, as the intrusive magmatic activity casues a local stress build up which causes the radial formation of dykes. The second tectonic activity, which results in the formation of the nearly parallel oriented dykes, is most likely caused by a regional extension, causing a tension large enough for the formation of the large parallel dyke systems.

5. Based on the patterns of the dyke systems, it is very likely that the first tectonic activity was the regional extension. After that the magmatic intrusions formed the radial oriented dykes.